

Test 3 - PEM Load-Limit Test

Purpose

The purpose of this test is to verify that the EMS only allows the PEM fuel cell to supply the load when the PEM operating conditions are met. Initially the Battery supplies the Load until SoC runs out.

The test should confirm that:

Condition	Expected scenario
HIGH price + no PV + battery unusable + PEM SoC > 10% + load < 40 mW	S6
HIGH price + no PV + battery unusable + load ≥ 40 mW	S1
HIGH price + no PV + battery unusable + PEM SoC ≤ 10%	S1

The main transition to observe is:

S6 ↔ **S1**, depending on whether the load is below or above **40 mW**.

Initial setup

Using Scripts:

AppLab: ems_base_estimation_logger_v5

Arduino IDE: ControllableLoad_V4.1_triggerAndData

Item	Value
Test duration	24 minutes
Simulated duration	24 hours
Price data	Real API day with HIGH prices
Load profile	Active, 15–75 mW
Initial Battery SoC	12%
Initial PEM SoC	80%
PV/lamp at start	Off

CSV logging	Enabled
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Demo and EMS control

Start the EMS system first. Then start the demo to reset the 96-slot timeline, begin CSV logging and activate the variable load trigger. Raw logging can run independently of the demo.

Reset

Start EMS system

Stop EMS system

Start demo + logging

Stop demo

Auto mode

Manual mode

Start raw log

Raw log

Stopped

Raw log file

-

Raw samples

0

Last raw update

-

Price test date

07/05/2026

Manual electricity price

0.99064

Hydrogen estimation (%)

80.0

Virtual battery charge state (%)

12.0

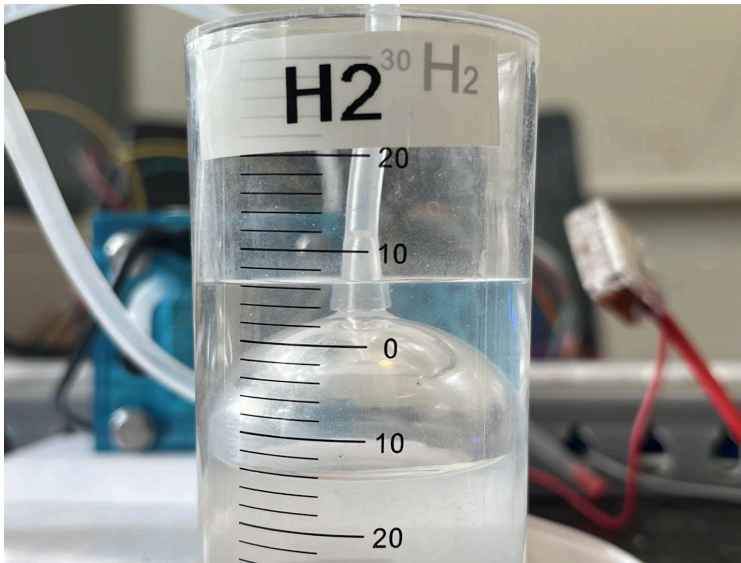
Load prices for date

Apply manual price

Initiate H2 estimation manually

Apply virtual battery state

Storage states		INA sensor measurements	
Current battery voltage	3.691 V	PV voltage	2.978 V
Initial SoC lookup voltage	3.693 V	Battery voltage	3.691 V
Lookup-estimated real battery SoC	4.6 %	PEM voltage	1.351 V
Integrated real battery SoC	4.7 %	Load voltage	5.122 V
Integrated real battery charge	93.353 mAh / 2000.0 mAh	PV current	0.10 mA
Virtual battery SoC	12.0 %	Battery current	-0.03 mA
Virtual battery capacity	10.00 mAh	PEM current	-0.04 mA
Virtual battery charge	1.200 mAh	Load current	-0.01 mA
Virtual battery state	low	PV power	0.30 mW
Hydrogen volume	14.00 mL	Battery power	-0.09 mW
Usable hydrogen	6.40 mL	PEM power	-0.06 mW
Hydrogen estimation	80.0 %	Load power	-0.04 mW
Hydrogen mode	idle	Load trigger	LOW
		Arduino mode	S1 Grid -> Load



Hydrogen Niveau før test (80% = 6.4mL)

Test procedure

1. Load the selected real API price day with HIGH prices.
2. Set the Battery virtual SoC below or at the unusable threshold.
3. Set the PEM virtual SoC above 10%.
4. Start the load profile.
5. Keep the lamp off for the entire test.
6. Start CSV logging.
7. Start the EMS test.
8. Observe whether the EMS selects **S6** when the load is below **40 mW**.
9. Observe whether the EMS selects **S1** when the load is equal to or above **40 mW**.
10. Confirm that PEM SoC only decreases during periods where **S6** is active.
11. If the PEM SoC reaches 10%, confirm that the EMS remains in or switches to **S1**.
12. Stop the test after 24 minutes.
13. Save the CSV file and screenshots.

CSV file

Item	Entry
CSV filename	/app/python/logs/ems_demo_20260512_142228.csv
Saved location	

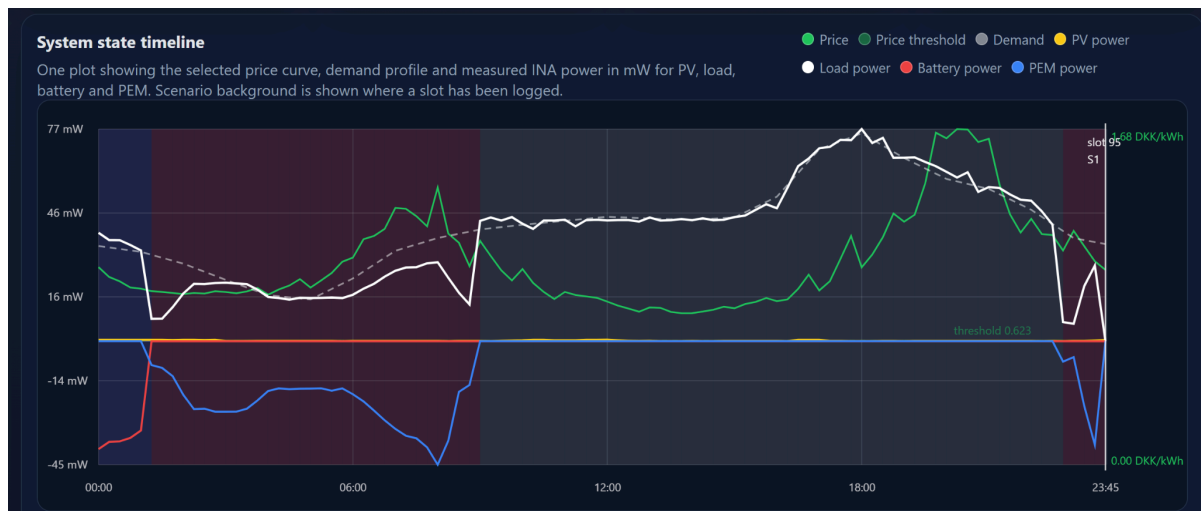
Notes

Time / timestep	Observation
00:00	Test start, S5 Battery to Load, initial SoC 12%
01:15	S6 PEM to Load is on, after battery is discharged to 10% It's very visible that once the switch has happened, the PEM takes a long time to reach steady state around the Demand setpoint
03:15	PEM reaches the demand setpoint. The PEM can only supply the demand setpoint whilst at the lowest point, afterwards it lies vastly below target.
09:00	PEM is switched off after demand <40mW S1 Grid to Load despite high prices. (Fallback) Both Batt and PEM conditions are not met.

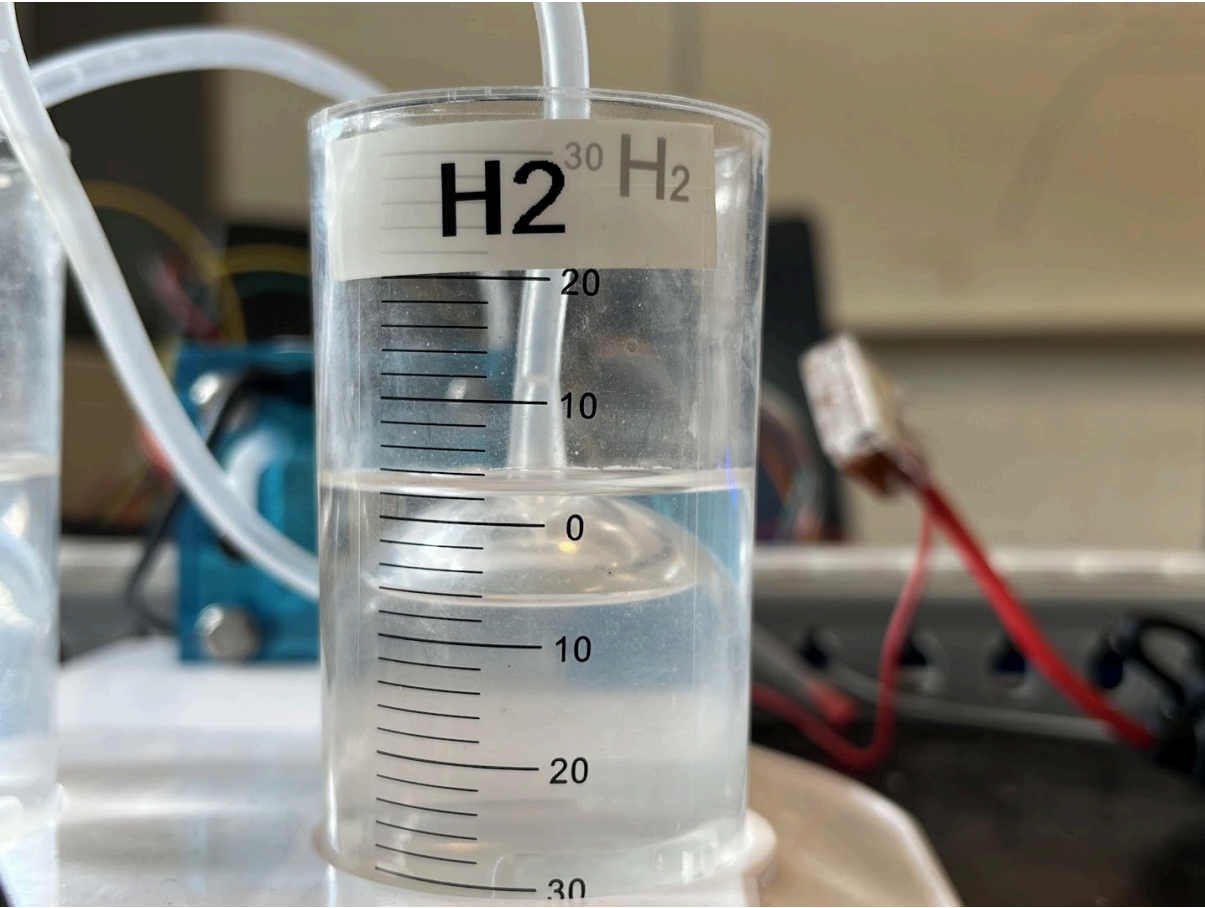
22:45	PEM is switched on again after Demand drops below 40mW. the PEM is unable to ramp up to the demand setpoint in the timespan.
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Result summary

Question	Answer
Did the system select S6 when load was below 40 mW?	Yes, after Battery Discharge
Did the system select S1 when load was ≥ 40 mW?	
Did PEM SoC decrease only during S6?	
Did the EMS switch to S1 when PEM SoC reached 10%, if reached?	
Were there rapid or unexpected switches around 40 mW?	
Any unexpected scenario changes?	
Overall result	Pass / Partial / Fail



Storage states		INA sensor measurements	
Current battery voltage	3.692 V	PV voltage	2.993 V
Initial SoC lookup voltage	3.693 V	Battery voltage	3.692 V
Lookup-estimated real battery SoC	4.6 %	PEM voltage	0.824 V
Integrated real battery SoC	4.7 %	Load voltage	5.122 V
Integrated real battery charge	93.136 mAh / 2000.0 mAh	PV current	0.10 mA
Virtual battery SoC	10.0 %	Battery current	-0.03 mA
Virtual battery capacity	10.00 mAh	PEM current	-0.04 mA
Virtual battery charge	0.999 mAh	Load current	-0.01 mA
Virtual battery state	empty	PV power	0.30 mW
Hydrogen volume	9.37 mL	Battery power	-0.09 mW
Usable hydrogen	1.77 mL	PEM power	-0.04 mW
Hydrogen estimation	22.1 %	Load power	-0.04 mW
Hydrogen mode	idle	Load trigger	LOW
		Arduino mode	S1 Grid -> Load



Hydrogen Niveau after Test (22.1% = 1.77mL)